

An explicit family of classical states refuting a conjecture of Linden and Winter

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Abstract

Linden and Winter (*Commun. Math. Phys.* 259:129–138, 2005) proved that any four-party quantum state with $I(A ; C|B) = I(C ; B|A) = I(A ; B|D) = 0$ satisfies $I(C ; D) \geq I(C ; AB)$, and conjectured (their Conjecture 6) that this constrained inequality is stable: that positive constants k_1, k_2, k_3 exist with $k_1 I(A ; C|B) + k_2 I(C ; B|A) + k_3 I(A ; B|D) + I(C ; D) - I(C ; AB) \geq 0$ for all quadripartite states. We exhibit an explicit one-parameter family of *classical* probability distributions on four bits — six atoms, integer weights in closed form — on which $I(A ; C|B)$ and $I(A ; B|D)$ vanish identically while the ratio $[I(C ; AB) - I(C ; D)]/I(C ; B|A)$ grows without bound. Consequently no choice of positive constants satisfies the conjectured inequality, even for classical states: the conjecture is false. The vanishing of the two conditional mutual informations is an exact combinatorial identity; the divergence of the ratio follows from an elementary asymptotic analysis, and is independently certified by directed-rounding interval arithmetic, which also yields explicit refuting states for every (k_1, k_2, k_3) with $k_2 \leq 10^{24}$. All certificates re-verify from a single dependency-free Python script.

1 Introduction

Which linear inequalities constrain the von Neumann entropies of the reduced states of a multipartite quantum state? For up to three parties the answer is classical: strong subadditivity and weak monotonicity [1] generate all inequalities, and Pippenger [2] asked whether anything new appears for four or more parties. No unconstrained inequality beyond the Lieb–Ruskai family has been found since 1973.

Linden and Winter [3] made the first advance past this point, proving a new *constrained* inequality:

Theorem (Linden–Winter). If a four-party state ρ_{ABCD} satisfies $I(A ; C|B) = I(C ; B|A) = I(A ; B|D) = 0$, then $I(C ; D) \geq I(C ; AB)$.

They closed the paper by conjecturing that the constraint can be relaxed linearly:

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Conjecture 6 (Linden–Winter). There exist positive real constants k_1, k_2, k_3 such that for all quadripartite quantum states,

$$k_1 I(A ; C|B) + k_2 I(C ; B|A) + k_3 I(A ; B|D) + I(C ; D) - I(C ; AB) \geq 0. \quad (1)$$

A proof would have produced the first new unconstrained inequality for the von Neumann entropy in over five decades. We show the conjecture fails, already for classical states.

Theorem 1. *No positive constants k_1, k_2, k_3 satisfy (1). Specifically, for the family W_L of classical distributions in Definition 2: $I(A ; C|B)$ and $I(A ; B|D)$ vanish identically on the family, and $R_L := [I(C ; AB) - I(C ; D)] / I(C ; B|A) \rightarrow \infty$ as $L \rightarrow \infty$. Hence for any positive (k_1, k_2, k_3) , every L with $R_L > k_2$ gives $f_k(W_L) < 0$ in (1).*

Classical distributions embed as diagonal quantum states, so a classical counterexample refutes the quantum conjecture. The constrained theorem itself is untouched: on the family, two of its three hypotheses hold exactly but the third, $I(C ; B|A) = 0$, fails — by an amount that shrinks too fast, which is precisely what the conjecture forbade. The result says the Linden–Winter inequality is not linearly stable: the deficit $I(C ; AB) - I(C ; D)$ can exceed any fixed multiple of the constraint violations.

Provenance. The family was found by an automated research system conducting a systematic search over supports and weight patterns, and the numerical certificates below were produced and cross-checked by four independently written implementations (no shared code), including the dependency-free verifier published with this note. The mathematics presented here — the identities of Lemma 3 and the asymptotics of Proposition 4 — is human-checkable by hand.

2 The family

Proposition 2 (Definition). *For an integer $L \geq 2$ let W_L be the probability distribution on $\{0, 1\}^4$ (bits ordered A, B, C, D) supported on six atoms, with weights proportional to:*

<i>atom</i> (A, B, C, D)	<i>weight</i>
$(0, 0, 0, 0)$	$2 \cdot 10^{4L+3}$
$(0, 0, 1, 0)$	$27 \cdot 10^{4L+3} - 27 \cdot 10^{3L+3}$
$(1, 1, 0, 1)$	$2 \cdot 10^{4L}$
$(1, 1, 1, 1)$	$27 \cdot 10^{4L}$
$(0, 1, 0, 1)$	$2 \cdot 10^{L+3}$
$(0, 1, 1, 0)$	$27 \cdot 10^{L+3}$

Equivalently, with $\delta = 10^{-L}$, the weights are proportional to $2, 27(1 - \delta), \frac{2}{1000}, \frac{27}{1000}, 2\delta^3, 27\delta^3$ on the same atoms.

Lemma 3 (Two exact zeros). *For every L : $I(A ; B|D) = 0$ and $I(A ; C|B) = 0$ on W_L , exactly.*

Proof. $I(A ; B|D)$: on the three atoms with $D = 0$ the bit A is constant ($= 0$), and on the three atoms with $D = 1$ the bit B is constant ($= 1$). Conditioned on either value of D , one of the two arguments of the mutual information is deterministic, so each conditional term vanishes.

$I(A ; C|B)$: on the two atoms with $B = 0$ the bit A is constant ($= 0$), so that block contributes zero. On the $B = 1$ block the joint distribution of (A, C) has the 2×2 weight matrix $\begin{pmatrix} w_{0101} & w_{0110} \\ w_{1101} & w_{1111} \end{pmatrix}$, and A and C are conditionally independent iff its determinant vanishes: $w_{1101} w_{0110} = w_{1111} w_{0101}$. Both products equal $54 \cdot 10^{5L+3}$. $\square$

Because of Lemma 3, on the family the left side of (1) collapses to

$$f_k(W_L) = k_2 y_L + e_L, \quad y_L := I(C ; B|A), \quad e_L := I(C ; D) - I(C ; AB), \quad (2)$$

for every (k_1, k_3) : those two coefficients multiply exact zeros.

3 Why the ratio diverges

Write $\delta = 10^{-L}$ and use the normalized form of Definition 2. Two structural observations drive everything.

Proposition 4. *As $\delta \rightarrow 0$: $y_L = \Theta(\delta^5)$ while $-e_L = \Theta(\delta^4)$, with $e_L < 0$ for all small δ . Consequently $R_L = -e_L/y_L = \Theta(1/\delta) \rightarrow \infty$.*

Proof sketch, elementary and self-contained. (i) y_L needs two rare events at once. Only the $A=0$ block contributes to $I(C ; B|A)$ (given $A=1$, $B \equiv 1$). Within $A=0$, the branch $B=1$ carries probability $\Theta(\delta^3)$ (the two δ^3 -atoms), and conditioned on the two branches the distributions of C are $(2, 27)/29$ versus $(2, 27(1 - \delta))/(29 - 27\delta)$ — total-variation distance $\Theta(\delta)$. A mutual information with a $\Theta(\delta^3)$ -rare branch whose conditional laws differ by $\Theta(\delta)$ is $\Theta(\delta^3) \cdot \Theta(\delta^2) = \Theta(\delta^5)$, by the local χ^2 -expansion of Kullback–Leibler divergence.

(ii) e_L vanishes exactly when the δ^3 -atoms are removed. Set the two δ^3 -weights to zero. Then only four atoms remain, on which the partition of the sample space induced by D coincides with the partition induced by (A, B) : the $D=0$ atoms are exactly the $AB=00$ atoms and the $D=1$ atoms are exactly the $AB=11$ atoms. Hence $I(C ; D) = I(C ; AB)$ identically in δ on the truncated family, i.e. $e \equiv 0$. Restoring the two atoms perturbs each side of e_L by $\Theta(\delta^3) \cdot \Theta(\delta) = \Theta(\delta^4)$: each δ^3 -atom sits in a D -block whose conditional law of C it shifts at relative scale $\Theta(\delta)$ — the atom $(0, 1, 1, 0)$ joins the $D=0$ block, where C 's law is δ -tilted, and the atom $(0, 1, 0, 1)$ joins the $D=1$ block, where it is the sole δ -scale disturbance. The two placements enter $I(C ; D)$ and $I(C ; AB)$ asymmetrically (under (A, B) both atoms sit in their own block $AB=01$, where together they leave C 's conditional law *exactly* equal to $(2, 27)/29$), and the $\Theta(\delta^4)$ mismatch has a strictly negative sign for small δ , which the interval certificates of Section 4 pin down together with its constant.

Combining (i) and (ii): $R_L = \Theta(\delta^4)/\Theta(\delta^5) = \Theta(1/\delta)$. $\square$

Remark 5. The certified computation below measures the sharp law: $R_L \delta \rightarrow 2$, i.e. $R_L = 2 \cdot 10^L - c + o(1)$ with $c \approx 0.6437$; and the per-step ratio R_{L+1}/R_L is certified ≥ 1.9 (measured: $\rightarrow 10$) at every step $2 \leq L \leq 15$. Theorem 1 needs only $R_L \rightarrow \infty$.

4 Machine certificates

The published verifier reproduces, from the integer weights alone, with directed-rounding decimal interval arithmetic (every logarithm bracketed one-sidedly; every certified sign a theorem about the printed digits):

L	certified lower bound on R_L
2	189.4773957
3	1,998.356506
4	19,999.25622
8	199,999,999.3
12	$1.999999999 \cdot 10^{12}$
16	$1.999999999 \cdot 10^{16}$

together with $y_L > 0$ and $e_L < 0$ for $2 \leq L \leq 16$, the per-step growth bound, and explicit kill certificates $f_k(W_L) < 0$ including

$$k = (1, 1, 1) \text{ at } L=2, \quad k = (10^9, 10^9, 10^9) \text{ at } L=15, \quad k = (10^{30}, 100, 10^{30}) \text{ at } L=5,$$

$$k = (0, 10^{20}, 0) \text{ at } L=22, \quad k = (0, 10^{24}, 0) \text{ at } L=26.$$

By (2) a certificate at k_2 covers all (k_1, k_2, k_3) , so every conjectured constant triple with $k_2 \leq 10^{24}$ is refuted by a named, explicit distribution; the verifier accepts an arbitrary triple and searches out the certifying L . Four independently written implementations (interval engines in independent codebases, no shared code) agree on every row.

5 Discussion

What survives, and what closes. The Linden–Winter constrained inequality (their Theorem 1) is untouched, as are the infinite constrained families of Cadney–Linden–Winter [5]. What closes is the named route from the constrained inequality to an unconstrained one: no linear relaxation of the constraints works. This is consistent with, and sharpens, the current expert assessment that no unconstrained non-Shannon-type inequality for the von Neumann entropy is in sight [6]: the one standing candidate is now gone, through states as simple as six-atom classical distributions on four bits.

The mechanism is reminiscent of the robustness failures of quantum Markov chains identified by Ibinson, Linden and Winter [4]: smallness of conditional mutual information does not control the quantities one would like it to control. Here the failure is sharper — the deficit $I(C ; AB) - I(C ; D)$ is not merely uncontrolled but exceeds *any* fixed linear multiple of the constraint violations.

References

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